

Euler's Identity and exponential form

Basic skills

Q1. a) i) $z = 2 + 3i$

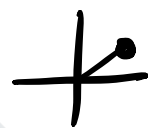
$$|z| = \sqrt{2^2 + 3^2} = \sqrt{13}$$

$$\text{Arg } z = \tan^{-1}\left(\frac{3}{2}\right) = 0.983$$

$$|z| = \sqrt{x^2 + y^2}$$

$$\text{Arg } z = \tan^{-1}\left(\frac{y}{x}\right)$$

careful if $x < 0$

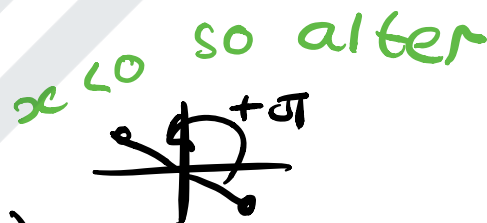


ii) $w = 3i - 2 = -2 + 3i$

$$|z| = \sqrt{(-2)^2 + 3^2} = \sqrt{13}$$

$$\text{Arg } z = \tan^{-1}\left(\frac{3}{-2}\right) = (-0.983)$$

$$\text{so } -0.983 + \pi$$



iii) $u = -z \leftarrow \text{could use } u = -2 - 3i$

$$|u| = |-z| = |z| = \sqrt{13}$$

$$\text{Arg } u = \text{Arg}(-z) = \text{Arg } z \pm \pi$$

depends on range

iv) $v = -w \leftarrow \text{or } v = 2 - 3i$ $= 0.983 - \pi$

$$|v| = |-w| = |w| = \sqrt{13}$$

$$\text{Arg } v = \text{Arg}(-w) = \text{Arg } w \pm \pi = -0.983$$

$$b) \quad z = \sqrt{13} (\cos(0.983) + i \sin(0.983))$$

$$w = \sqrt{13} (\cos(\pi - 0.983) + i \sin(\pi - 0.983))$$

$$u = \sqrt{13} (\cos(0.983 - \pi) + i \sin(0.983 - \pi))$$

$$v = \sqrt{13} (\cos(-0.983) + i \sin(-0.983))$$

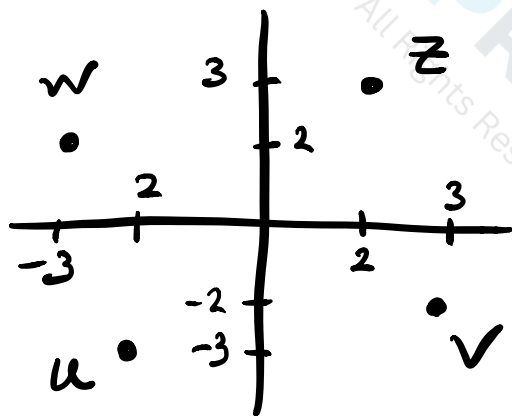
$$c) \quad z = \sqrt{13} e^{0.983i}$$

$$w = \sqrt{13} e^{(\pi - 0.983)i}$$

$$u = \sqrt{13} e^{(0.983 - \pi)i}$$

$$v = \sqrt{13} e^{-0.983i}$$

d)



Competency skills

Q2. By Euler's identity $e^{i\theta} = \cos\theta + i\sin\theta$

$$\text{So } e^{i\pi} = \cos\pi + i\sin\pi = -1 + 0i$$

$$\text{So } 1 + e^{i\pi} = 0$$

Q3. a) $2e^{\frac{\pi}{3}i}$ convert by mod-arg form

$$= 2 \left(\cos\left(\frac{\pi}{3}\right) + i \sin\left(\frac{\pi}{3}\right) \right)$$

$$= 1 + \sqrt{3}i$$

b) πe^{2i}

$$= \pi \left(\cos(2) + i \sin(2) \right)$$

$$= -1.307 + 2.857i$$

c) $\frac{1}{3}e^{-\frac{\pi}{8}i}$

$$= \frac{1}{3} \left(\cos\left(-\frac{\pi}{8}\right) + i \sin\left(-\frac{\pi}{8}\right) \right)$$

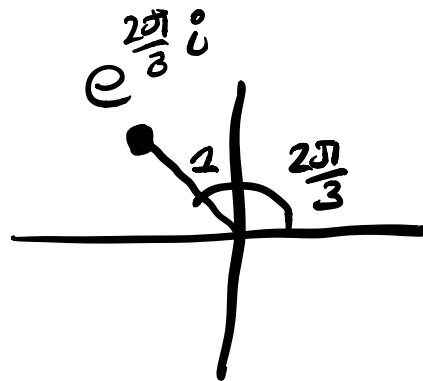
$$= 0.2697 - 0.1959i$$

Q4. a) $1e^{\frac{2\pi}{3}i}$

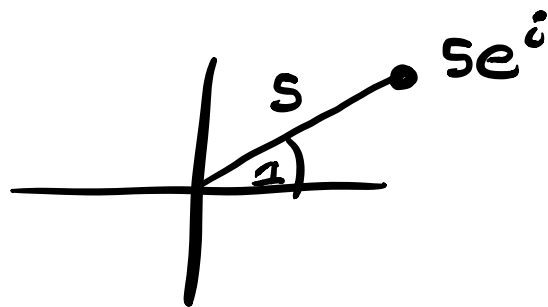
modulus
1

Argument
 $\frac{2\pi}{3}$

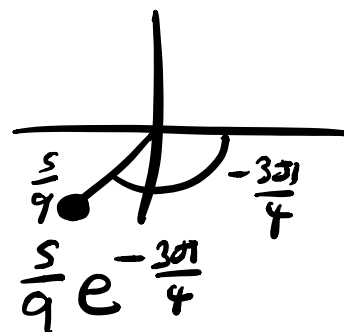
Form $|z|e^{\text{Arg } z i}$



b) $5e^{1i}$
 modulus 5 Argument 1



c) $\frac{5}{9}e^{-\frac{3\pi}{4}i}$
 modulus $\frac{5}{9}$
 Argument $-\frac{3\pi}{4}$



Q5. $z = 3e^{\frac{\pi}{3}i}$ $w = 4e^{-\frac{\pi}{2}i}$

a) $zw = 3e^{\frac{\pi}{3}i} \times 4e^{-\frac{\pi}{2}i} = 12e^{-\frac{\pi}{12}i}$
 Add arguments
 multiply moduli

b) $\frac{z}{w} = \frac{3e^{\frac{\pi}{3}i}}{4e^{-\frac{\pi}{2}i}} = \frac{3}{4}e^{\frac{2\pi}{12}i}$

c) $z + w = 3(\cos(\frac{\pi}{3}) + i\sin(\frac{\pi}{3})) + 4(\cos(-\frac{\pi}{2}) + i\sin(-\frac{\pi}{2}))$
 TO add Normal form is needed
 $= 1.5 + \frac{3\sqrt{3}}{2}i + 0 - 4i$

$$= 1.5 - \frac{8 - 3\sqrt{3}}{2} i$$

$$d) z - w = 1.5 + \frac{3\sqrt{3}}{2} i - (0 - 4i)$$

need
normal
form

$$= 1.5 + \frac{8 + 3\sqrt{3}}{2} i$$

Advanced skills

Q7. $z = -5e^{\frac{5\pi}{2}} i$

modulus of -5 Argument $\frac{5\pi}{2} = \frac{\pi}{2}$

modulus needs to be positive so issue here

so no issue with

-5 ? Argument

not possible

Q8. $z = 3 - 4i$

a) z^3 by $|z| = \sqrt{3^2 + 4^2} = 5$

$$\text{Arg } z = \tan^{-1}\left(\frac{-4}{3}\right) = -0.927$$

$$z = 5e^{-0.927i}$$

$$z^3 = 5^3 e^{-2.78i}$$

could use

$$(3 - 4i)^3$$

b) $z^{10} + 2i - 5$

$$z^{10} = 5^{10} e^{-9.27i} = 9765625 e^{(2\pi - 9.27)i}$$

$$= 9765625 (\cos(2\pi - 9.27) + i \sin(2\pi - 9.27))$$

$$= -9648884 - 1505475i$$

$$z^{10} + 2i - 5$$

$$= -964889 - 1505473i$$

c) $z^5 (2+i)^4 (1-i)^{-2}$

$$z^5 = 5^5 e^{-4.635i}$$

$$(2+i)^4 = (\sqrt{5} e^{0.464i})^4 = 25 e^{1.856i}$$

$$(1-i)^{-2} = (\sqrt{2} e^{-\frac{\pi}{4}i})^{-2} = 2 e^{-\frac{\pi}{2}i}$$

$$z^5 (2+i)^4 (1-i)^{-2} = 5^5 e^{-4.635i} \times 25 e^{1.856i} \times 2 e^{-\frac{\pi}{2}i} = 78125 e^{-4.35i}$$

Q9. Exponential form (like modulus-argument form) is useful when multiplying or dividing. However it is less useful when adding or subtracting